

Step 10 - Production-ize the Pipeline

Phase C · Parameters · Backfill · Trigger · Alerts · ~20 min

Goal: Production-ize: business_date parameter with backfill proven, 7 PM ET schedule trigger, 2x retry, Azure Monitor failure alert - then stop the trigger for cost hygiene. Artifacts: pipeline JSON v2 + trigger JSON + screenshots.

LEARN

Why utcnow() is not enough: EOD fails at 3 AM, you rerun at 9 AM FOR YESTERDAY. A hardcoded date cannot backfill. Parameter pattern: empty = today, value = backfill. That is idempotent, parameterized design - interviewers probe it.

Parameters vs variables: parameters are read-only inputs from outside (trigger, UI, parent pipeline); variables are internal scratch a pipeline can reassign via Set Variable.

ADF quirk: parameter defaults are static text - you cannot default to @utcnow(). Standard pattern: default empty + an @if downstream.

Retry before alert: transient blips should not page anyone. 2 retries, 30s, THEN fail loudly.

LAB

10.1 — Dataset parameter: open DS_Bronze_Transactions - Parameters - New: business_date (String). Connection tab Directory expression becomes:

```
@concat('transactions/business_date=', dataset().business_date)
```

10.2 — Pipeline parameter + logic: pipeline canvas - Parameters - New: business_date, default EMPTY. Copy activity - Sink tab - the new business_date field - Add dynamic content:

```
@if(equals(pipeline().parameters.business_date, ''),
    formatDateTime(utcnow(), 'yyyy-MM-dd'),
    pipeline().parameters.business_date)
```

10.3 — Retry: Copy activity - General - Retry 2, interval 30s.

10.4 — Prove backfill: Debug - prompted for business_date - enter 2025-11-15. After success, Storage browser shows TWO folders: today's AND business_date=2025-11-15. Screenshot it - that is rerun-ability, proven.

10.5 — Schedule trigger: Add trigger - New: TR_Daily_EOD_7PM, Schedule, every 1 day, time zone Eastern Time (US & Canada), execute 19:00, Activated. Leave the run's business_date EMPTY (= today via your @if). Then PUBLISH ALL - debug runs work unpublished, but triggers only fire on published pipelines (classic gotcha).

10.6 — Trigger now - then Monitor (gauge) - Pipeline runs: see Triggered-by Manual trigger vs your earlier Debug runs.

10.7 — Failure alert: Monitor - Alerts & metrics - New alert rule: ALERT_Pipeline_Failed, Sev1, metric Failed pipeline runs greater than 0, action group with your email. The minimum-viable on-call setup.

10.8 — Cost hygiene: Manage - Triggers - TR_Daily_EOD_7PM - STOP. Config stays; screenshots + JSON are the evidence; a daily learning run serves nobody.

COMMIT + CHECKPOINT

To adf/: updated pipeline JSON (parameters + @if), trigger JSON, screenshots (two business_date folders, trigger at 19:00 Eastern, Monitor run, alert rule). README line: 'Parameterized by business_date (empty=today, value=backfill), daily 7 PM ET trigger, 2x retry, failure alerting.' Phase C complete.

Interview line you have earned: "ADF orchestrates ingestion to the bronze layer with business_date parameterization and proven backfill; Databricks processes Bronze to Silver to Gold with PII masking and FINTRAC reporting."

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